

MotoGP Dissertation

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MotoGP

```
MotoGP2025_Race <- read_table("MotoGP2025_Race.txt")
```

```
##  
## -- Column specification -----  
## cols(  
##   .default = col_character()  
## )  
## i Use 'spec()' for the full column specifications.
```

```
MotoGP2025_Sprint <- read_table("MotoGP2025_Sprint.txt")
```

```
##  
## -- Column specification -----  
## cols(  
##   .default = col_character()  
## )  
## i Use 'spec()' for the full column specifications.
```

```
MotoGP2024_Race <- read_table("MotoGP2024_Race.txt")
```

```
##  
## -- Column specification -----  
## cols(  
##   .default = col_character()  
## )  
## i Use 'spec()' for the full column specifications.
```

```
MotoGP2024_Sprint <- read_table("MotoGP2024_Sprint.txt")
```

```
##  
## -- Column specification -----  
## cols(  
##   .default = col_character()  
## )  
## i Use 'spec()' for the full column specifications.
```

```
MotoGP2023_Race <- read_table("MotoGP2023_Race.txt")
```

```
##
## -- Column specification -----
## cols(
##   .default = col_character()
## )
## i Use 'spec()' for the full column specifications.
```

```
MotoGP2023_Sprint <- read_table("MotoGP2023_Sprint.txt")
```

```
##
## -- Column specification -----
## cols(
##   .default = col_character()
## )
## i Use 'spec()' for the full column specifications.
```

MotoGP2025

Sprint Race 25

```
head(MotoGP2025_Sprint)
```

```
## # A tibble: 6 x 25
##   Rider Bike Team S.THA S.ARG S.AME S.QAT S.SPA S.FRA S.GBR S.ARA S.ITA S.NED
##   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 M.Mar~ Duca~ Duca~ 1     1     1     1     1     1     2     1     1     1
## 2 A.Mar~ Duca~ BK8G~ 2     2     2     2     2     2     1     2     2     2
## 3 Bezze~ Apri~ Apri~ 12    6    10     9     8    17     4     8     6     3
## 4 Acosta KTM  RedB~ 6     9     7    11    10    19     8     5    Ret     9
## 5 Bagna~ Duca~ Duca~ 3     3     3     8     3    NC     6    12     3     5
## 6 Di_Gi~ Duca~ Pert~ Ret    5     4     6     6     7     3     6     5     4
## # i 12 more variables: S.GER <chr>, S.CZE <chr>, S.AUT <chr>, S.HUN <chr>,
## #   S.CAT <chr>, S.RSM <chr>, S.JPN <chr>, S.INA <chr>, S.AUS <chr>,
## #   S.MAL <chr>, S.POR <chr>, S.VAL <chr>
```

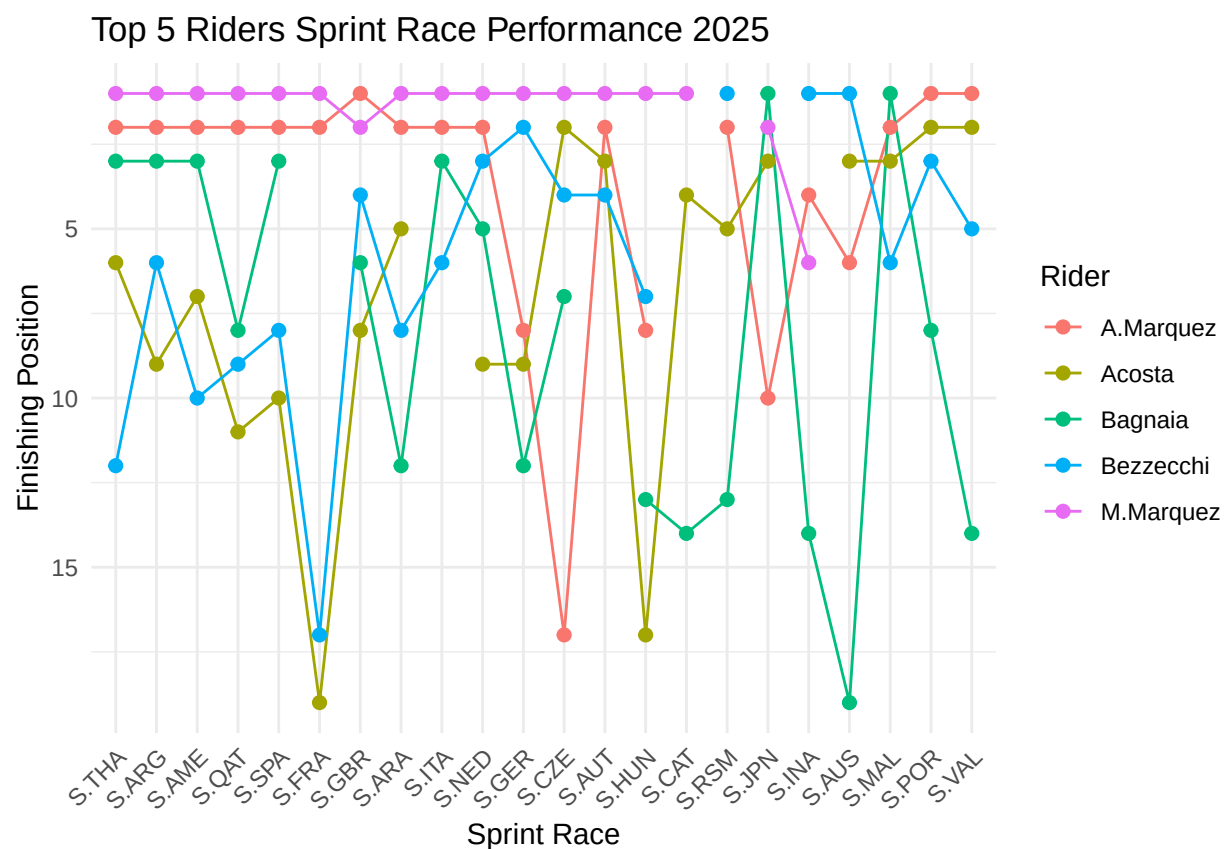
```
top5_sprint <- MotoGP2025_Sprint %>%
  slice(1:5) %>%
  pivot_longer(
    cols = S.THA:S.VAL,
    names_to = "Sprint_Race",
    values_to = "Result"
  ) %>%
  mutate(
    Position = as.numeric(Result),
    Sprint_Race = factor(Sprint_Race, levels = names(MotoGP2025_Sprint)[4:25])
  )
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Position = as.numeric(Result)'.
## Caused by warning:
## ! NA introdotti per coercizione
```

```
ggplot(top5_sprint, aes(x = Sprint_Race, y = Position, colour = Rider, group = Rider)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_reverse() +
  labs(
    title = "Top 5 Riders Sprint Race Performance 2025",
    x = "Sprint Race",
    y = "Finishing Position",
    colour = "Rider"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## ('geom_line()').
```

```
## Warning: Removed 12 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Race 2025

```
head(MotoGP2025_Race)
```

```
## # A tibble: 6 x 25
##   Rider  Bike  Team  THA  ARG  AME  QAT  SPA  FRA  GBR  ARA  ITA  NED
##   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 M_Mar~ Duca~ Duca~ 1    1    Ret  1    12   2    3    1    1    1
## 2 A_Mar~ Duca~ BK8G~ 2    2    2    6    1   Ret  5    2    2   Ret
## 3 Bezze~ Apri~ Apri~ 6    Ret  6    9   14   14   1    8    5    2
## 4 Acosta KTM   RedB~ 19   8    Ret  8    7    4    6    4    8    4
## 5 Bagna~ Duca~ Duca~ 3    4    1    2    3   16   Ret  3    4    3
## 6 Di_Gi~ Duca~ Pert~ 10   5    3   16   5    8    9    9    3    6
## # i 12 more variables: GER <chr>, CZE <chr>, AUT <chr>, HUN <chr>, CAT <chr>,
## #   RSM <chr>, JPN <chr>, INA <chr>, AUS <chr>, MAL <chr>, POR <chr>, VAL <chr>
```

```
top5_race <- MotoGP2025_Race %>%
  slice(1:5) %>%
  pivot_longer(
    cols = THA:VAL,
    names_to = "Race",
    values_to = "Result"
  ) %>%
  mutate(
    Position = as.numeric(Result),
    Race = factor(Race, levels = names(MotoGP2025_Race)[4:25])
  )
```

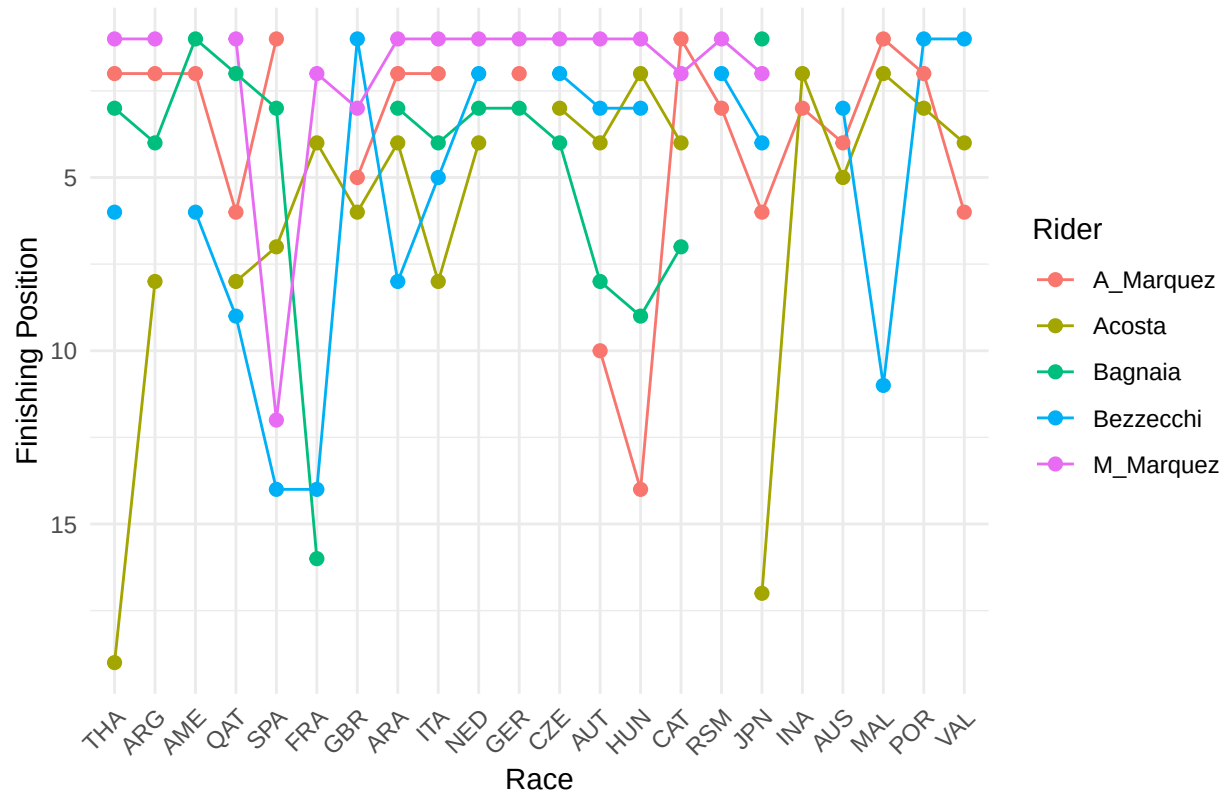
```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Position = as.numeric(Result)'.
## Caused by warning:
## ! NA introdotti per coercizione
```

```
ggplot(top5_race, aes(x = Race, y = Position, colour = Rider, group = Rider)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_reverse() +
  labs(
    title = "Top 5 Riders Race Performance 2025",
    x = "Race",
    y = "Finishing Position",
    colour = "Rider"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_line()').
```

```
## Warning: Removed 23 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Top 5 Riders Race Performance 2025



Analysis of the race 2025

```

moto_2025<-read.table("GP_2025.txt")
M_M <- moto_2025[1:20,1:36]

```

```

L2025M<-ordertable2supp(M_M)
head(L2025M)

```

```
## Warning in print.hyper2(x): powers have nonzero sum
```

```

## log(A_Marquez^32 * (A_Marquez + Acosta + Aldeguer + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Giannantonio + M_Marquez + Marini
## + Miller + Mir + Morbidelli + Ogura + Oliveira + Quartararo +
## R.Fernandez + Rins + Vinales)^-1 * (A_Marquez + Acosta + Aldeguer +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio + M_Marquez
## + Marini + Miller + Mir + Morbidelli + Ogura + Oliveira + Quartararo +
## R.Fernandez + Rins + Vinales + Zarco)^-36 * (A_Marquez + Acosta +
## Aldeguer + Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio
## + Marini + Miller + Mir + Morbidelli + Ogura + Oliveira + Quartararo +
## R.Fernandez + Rins + Vinales)^-1 * (A_Marquez + Acosta + Aldeguer +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio + Marini +
## Miller + Mir + Morbidelli + Ogura + Oliveira + Quartararo + R.Fernandez

```

```
## + Rins + Vinales + Zarco)^-25 * (A_Marquez + Acosta + Aldeguer +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio + Marini +
## Miller + Mir + Morbidelli + Ogura + Oliveira + R.Fernandez + Rins +
## Vinales + Zarco)^-1)
```

```
L2025<- ordertable2supp(moto_2025[2:20,-ncol(moto_2025)])
head(L2025)
```

```
## Warning in print.hyper2(x): powers have nonzero sum
```

```
## log(A_Marquez^40 * (A_Marquez + Acosta + Aldeguer + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Giannantonio + Marini + Miller +
## Mir + Morbidelli + Ogura + Oliveira + Quartararo + R.Fernandez + Rins +
## Vinales)^-1 * (A_Marquez + Acosta + Aldeguer + Bagnaia + Bastianini +
## Bezzecchi + Binder + Di_Giannantonio + Marini + Miller + Mir +
## Morbidelli + Ogura + Oliveira + Quartararo + R.Fernandez + Rins +
## Vinales + Zarco)^-44 * (A_Marquez + Acosta + Aldeguer + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Giannantonio + Marini + Miller +
## Mir + Morbidelli + Ogura + Oliveira + Quartararo + Rins + Vinales +
## Zarco)^-1 * (A_Marquez + Acosta + Aldeguer + Bagnaia + Bastianini +
## Bezzecchi + Binder + Di_Giannantonio + Marini + Miller + Mir +
## Morbidelli + Ogura + Oliveira + R.Fernandez + Rins + Vinales +
## Zarco)^-1 * (A_Marquez + Acosta + Aldeguer + Bagnaia + Bastianini +
## Bezzecchi + Binder + Marini + Miller + Mir + Morbidelli + Ogura +
## Oliveira + Quartararo + R.Fernandez + Rins + Vinales + Zarco)^-1)
```

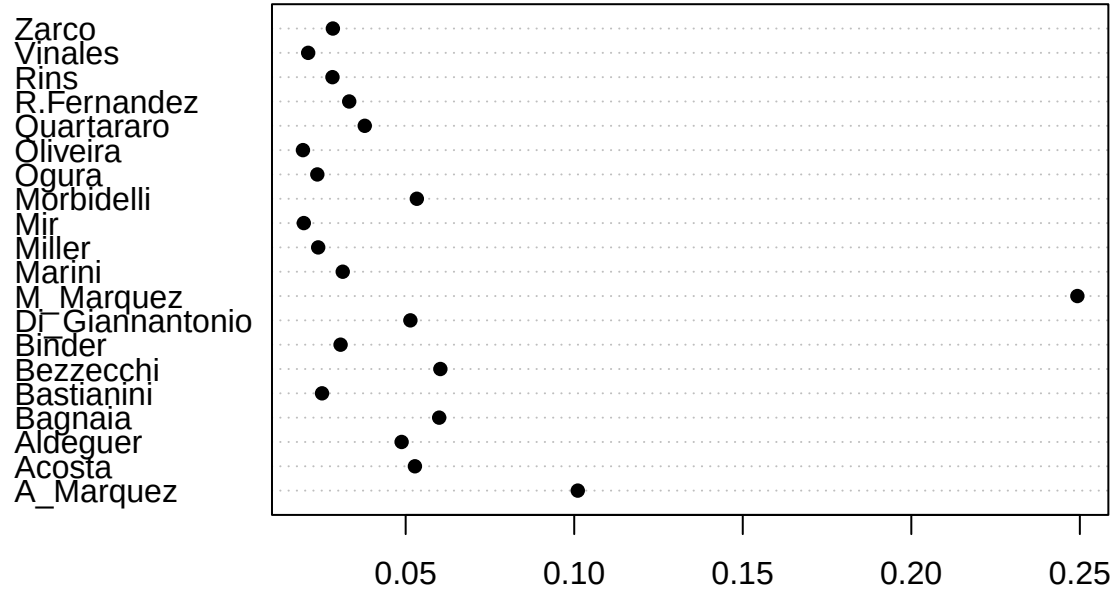
```
moto_maxp_M <- maxp(L2025M)
moto_maxp_M
```

##	A_Marquez	Acosta	Aldeguer	Bagnaia	Bastianini
##	0.10105134	0.05274240	0.04879676	0.05994217	0.02518596
##	Bezzecchi	Binder	Di_Giannantonio	M_Marquez	Marini
##	0.06029291	0.03068836	0.05139392	0.24926567	0.03132580
##	Miller	Mir	Morbidelli	Ogura	Oliveira
##	0.02404594	0.01977481	0.05330976	0.02379134	0.01952662
##	Quartararo	R.Fernandez	Rins	Vinales	Zarco
##	0.03785900	0.03324970	0.02828573	0.02107428	0.02839754

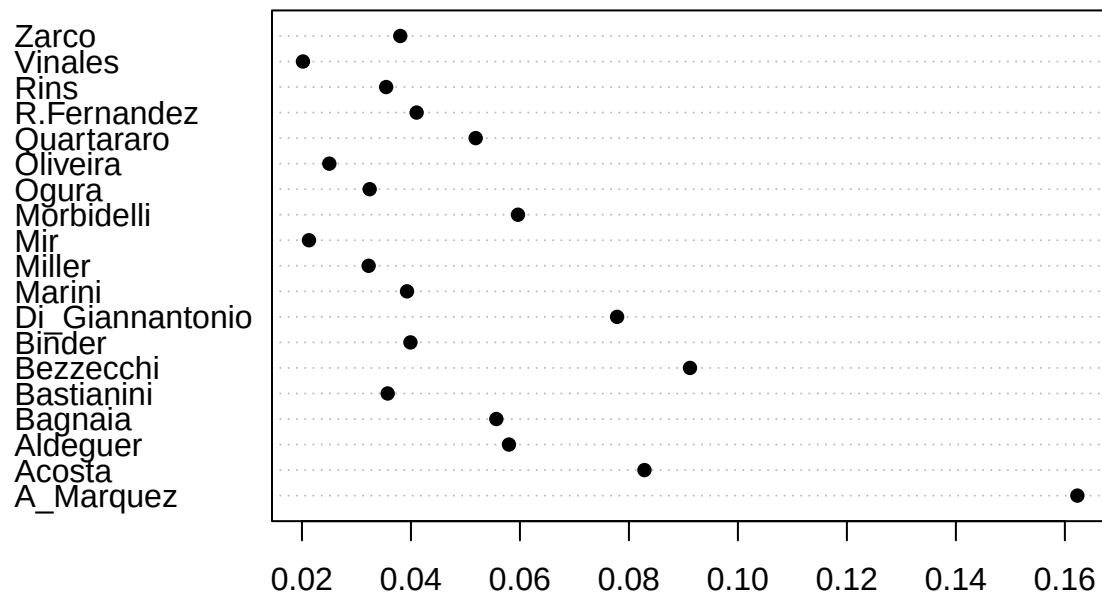
```
moto_maxp <- maxp(L2025)
moto_maxp
```

##	A_Marquez	Acosta	Aldeguer	Bagnaia	Bastianini
##	0.16229470	0.08285260	0.05798659	0.05566996	0.03573345
##	Bezzecchi	Binder	Di_Giannantonio	Marini	Miller
##	0.09120817	0.03990859	0.07783418	0.03928343	0.03223822
##	Mir	Morbidelli	Ogura	Oliveira	Quartararo
##	0.02128683	0.05964353	0.03242760	0.02502194	0.05188176
##	R.Fernandez	Rins	Vinales	Zarco	
##	0.04104402	0.03545716	0.02018430	0.03804295	

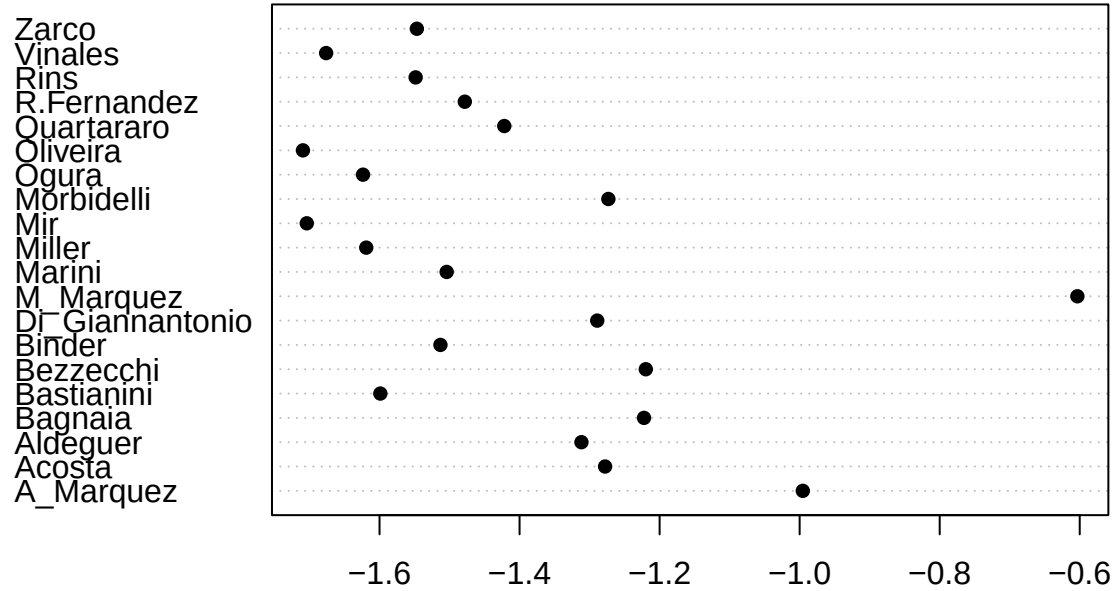
```
dotchart(moto_maxp_M, pch=16)
```



```
dotchart(moto_maxp, pch=16)
```

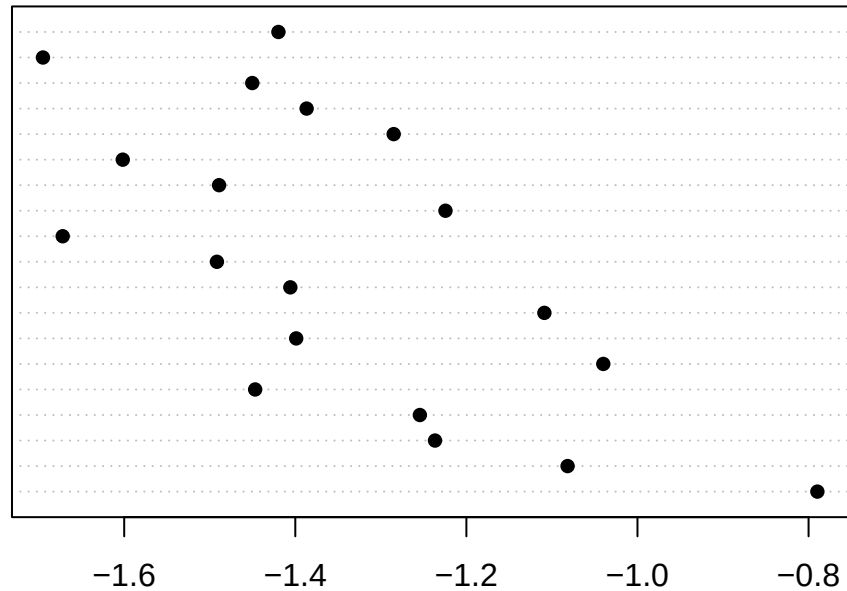


```
dotchart(log10(moto_maxp_M),pch=16)
```



```
dotchart(log10(moto_maxp),pch=16)
```

Zarco
Vinales
Rins
R.Fernandez
Quartararo
Oliveira
Ogura
Morbidei
Mir
Miller
Marini
Di Giannantonio
Binder
Bezzecchi
Bastianini
Bagnaia
Aldeguer
Acosta
A_Marquez



MotoGP 2024

```
head(MotoGP2024_Sprint)
```

```
## # A tibble: 6 x 23
##   Rider Bike Team S.QAT S.POR S.AME S.SPA S.FRA S.CAT S.ITA S.NED S.GER S.GBR
##   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Martin Duca~ Prim~ 1 3 3 1 1 4 Ret 2 1 1 2
## 2 Bagna~ Duca~ Duca~ 4 4 8 Ret Ret Ret 1 1 3 Ret
## 3 M.Mar~ Duca~ Gres~ 5 2 2 6 2 2 2 Ret 6 Ret
## 4 Basti~ Duca~ Duca~ 6 6 6 Ret 4 5 Ret 4 4 1
## 5 Binder KTM RedB~ 2 Ret 12 Ret 15 Ret 6 6 8 4
## 6 Acosta KTM RedB~ 8 7 4 2 6 3 3 10 22 5
## # i 10 more variables: S.AUT <chr>, S.ARA <chr>, S.RSM <chr>, S.EMI <chr>,
## # S.INA <chr>, S.JPN <chr>, S.AUS <chr>, S.THA <chr>, S.MAL <chr>,
## # S.SLD <chr>
```

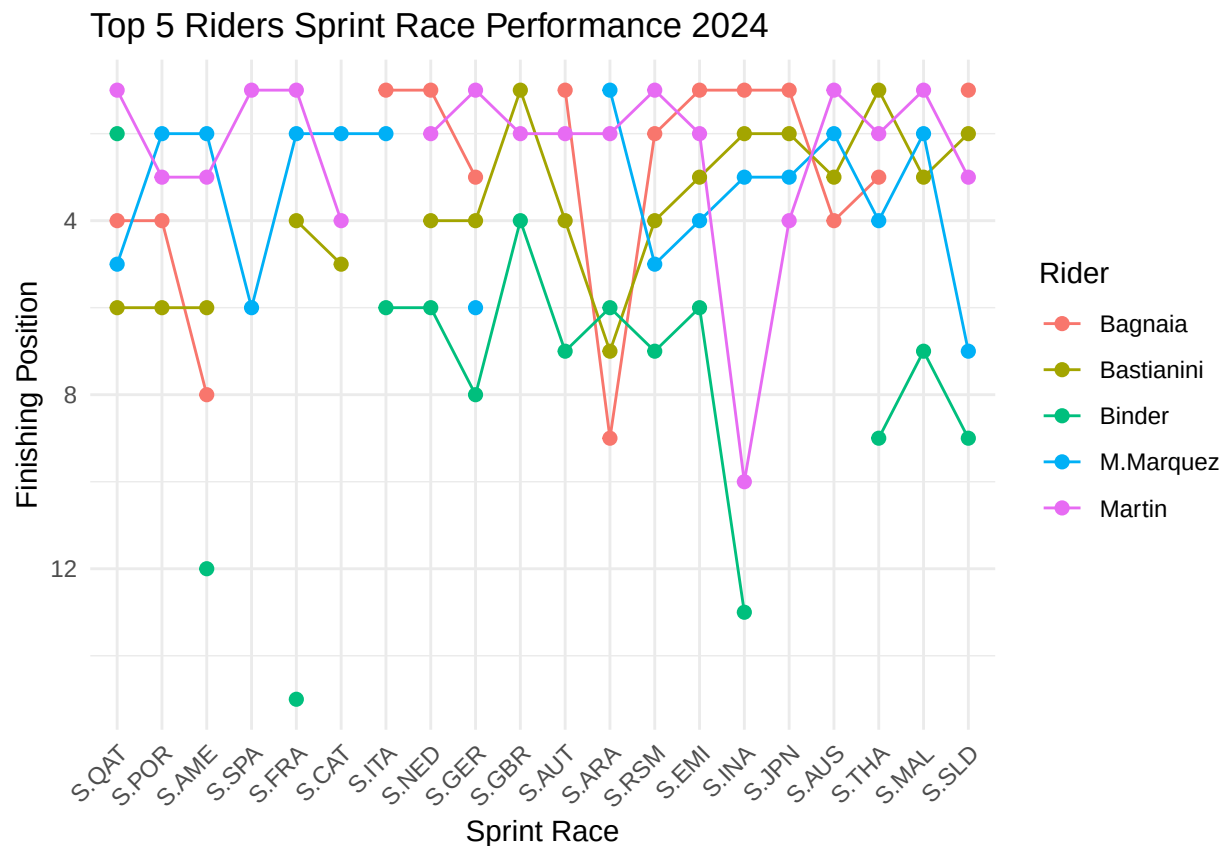
```
top5_sprint24 <- MotoGP2024_Sprint %>%
  slice(1:5) %>%
  pivot_longer(
    cols = S.QAT:S.SLD,
    names_to = "Sprint_Race24",
    values_to = "Result"
  ) %>%
  mutate(
    Position = as.numeric(Result),
```

```
Sprint_Race24 = factor(Sprint_Race24, levels = names(MotoGP2024_Sprint)[4:23])
)
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Position = as.numeric(Result)'.
## Caused by warning:
## ! NA introdotti per coercizione
```

```
ggplot(top5_sprint24, aes(x = Sprint_Race24, y = Position, colour = Rider, group = Rider)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_reverse() +
  labs(
    title = "Top 5 Riders Sprint Race Performance 2024",
    x = "Sprint Race",
    y = "Finishing Position",
    colour = "Rider"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Removed 16 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
head(MotoGP2024_Race)
```

```
## # A tibble: 6 x 23
##   Rider Bike Team QAT POR AME SPA FRA CAT ITA NED GER GBR
##   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Martin Duca~ Prim~ 3 1 4 Ret 1 2 3 2 Ret 2
## 2 Bagna~ Duca~ Duca~ 1 Ret 5 1 3 1 1 1 1 3
## 3 M.Mar~ Duca~ Gres~ 4 16 Ret 2 2 3 4 10 2 4
## 4 Basti~ Duca~ Duca~ 5 2 3 5 4 18 2 3 4 1
## 5 Binder KTM RedB~ 2 4 9 6 8 8 10 6 9 Ret
## 6 Acosta KTM RedB~ 9 3 2 10 Ret 13 5 Ret 7 9
## # i 10 more variables: AUT <chr>, ARA <chr>, RSM <chr>, EMI <chr>, INA <chr>,
## # JPN <chr>, AUS <chr>, THA <chr>, MAL <chr>, SLD <chr>
```

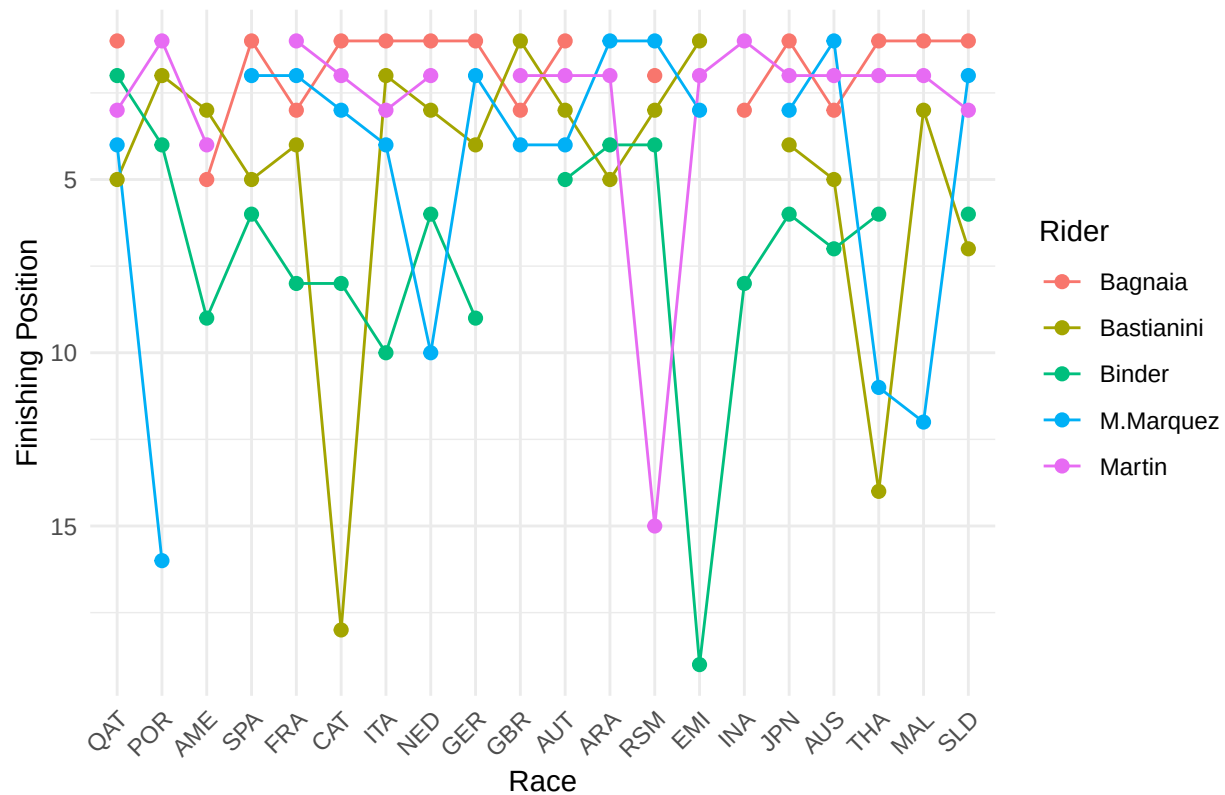
```
top5_race24 <- MotoGP2024_Race %>%
  slice(1:5) %>%
  pivot_longer(
    cols = QAT:SLD,
    names_to = "Race24",
    values_to = "Result"
  ) %>%
  mutate(
    Position = as.numeric(Result),
    Race24 = factor(Race24, levels = names(MotoGP2024_Race)[4:23])
  )
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Position = as.numeric(Result)'.
## Caused by warning:
## ! NA introdotti per coercizione
```

```
ggplot(top5_race24, aes(x = Race24, y = Position, colour = Rider, group = Rider)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_reverse() +
  labs(
    title = "Top 5 Riders Race Performance 2024",
    x = "Race",
    y = "Finishing Position",
    colour = "Rider"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Top 5 Riders Race Performance 2024



```
moto_2024<-read.table("GP_2024.txt")
```

```
L2024<- ordertable2supp(moto_2024[1:20,-ncol(moto_2024)])
head(L2024)
```

```
## Warning in print.hyper2(x): powers have nonzero sum
```

```
## log(A_Espargaro^30 * (A_Espargaro + A_Fernandez + A_Marquez + Acosta +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio + M_Marquez
## + Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Rins + Vinales + Zarco)^-40 * (A_Espargaro + A_Fernandez
## + A_Marquez + Acosta + Bagnaia + Bastianini + Bezzecchi + Binder +
## Di_Giannantonio + M_Marquez + Martin + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Zarco)^-3 * (A_Espargaro +
## A_Fernandez + A_Marquez + Acosta + Bagnaia + Bastianini + Bezzecchi +
## Binder + Di_Giannantonio + M_Marquez + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Vinales + Zarco)^-10 *
## (A_Espargaro + A_Fernandez + A_Marquez + Acosta + Bagnaia + Bastianini
## + Bezzecchi + Binder + Di_Giannantonio + M_Marquez + Miller +
## Morbidelli + Nakagami + Quartararo + R_Fernandez + Rins + Vinales +
## Zarco)^-1 * (A_Espargaro + A_Fernandez + A_Marquez + Acosta + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Giannantonio + Martin + Miller +
## Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez + Rins +
## Vinales + Zarco)^-4)
```

```

moto_maxp24 <- maxp(L2024)
moto_maxp24

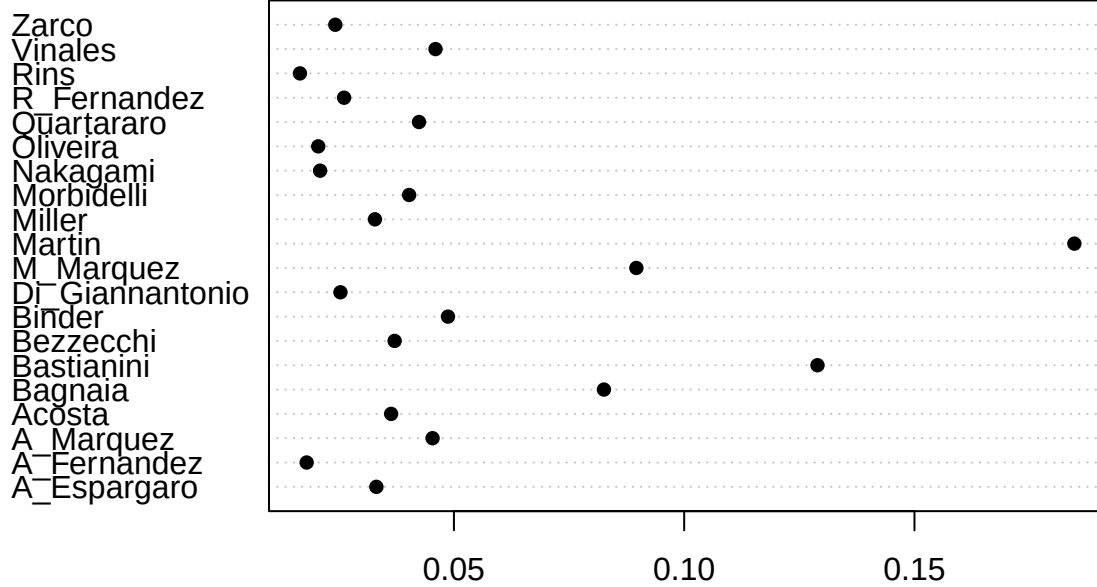
```

##	A_Espargaro	A_Fernandez	A_Marquez	Acosta	Bagnaia
##	0.03316597	0.01802943	0.04536336	0.03637145	0.08256965
##	Bastianini	Bezzecchi	Binder	Di_Giannantonio	M_Marquez
##	0.12894345	0.03713081	0.04871255	0.02536275	0.08961565
##	Martin	Miller	Morbidelli	Nakagami	Oliveira
##	0.18472174	0.03285226	0.04027684	0.02094843	0.02054235
##	Quartararo	R_Fernandez	Rins	Vinales	Zarco
##	0.04242150	0.02614924	0.01657044	0.04600430	0.02424784

```

dotchart(moto_maxp24,pch=16)

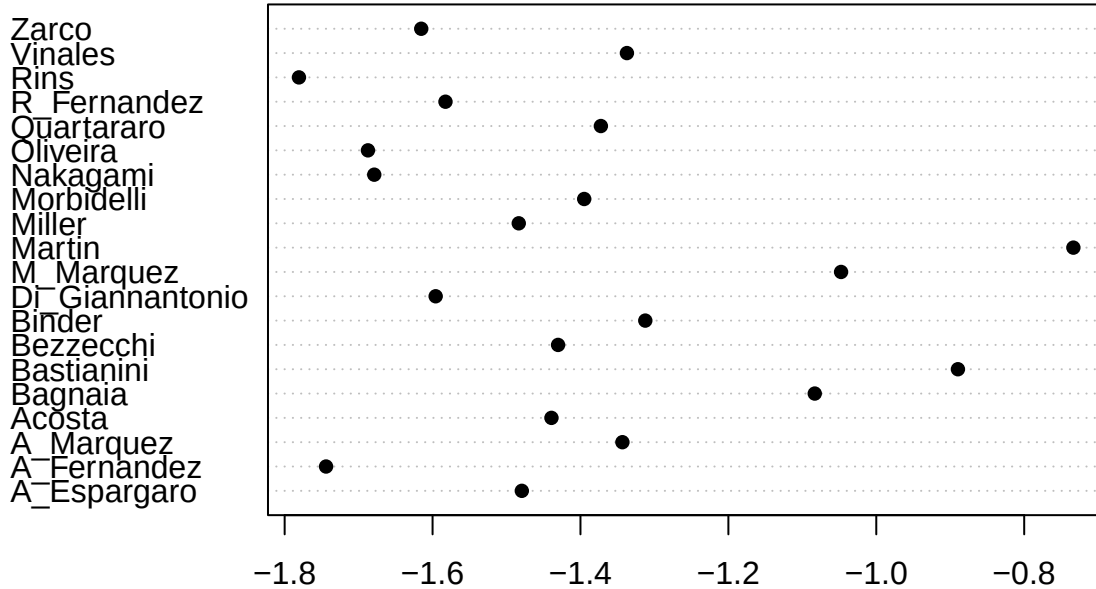
```



```

dotchart(log10(moto_maxp24),pch=16)

```



```
Sprint24 <-read.table("Sprint_2024.txt")

L_Sprint24 <- ordertable2supp(Sprint24[1:20,])
head(L_Sprint24)
```

Difference in Rider Performance Between Sprint and Race

```
## Warning in print.hyper2(x): powers have nonzero sum

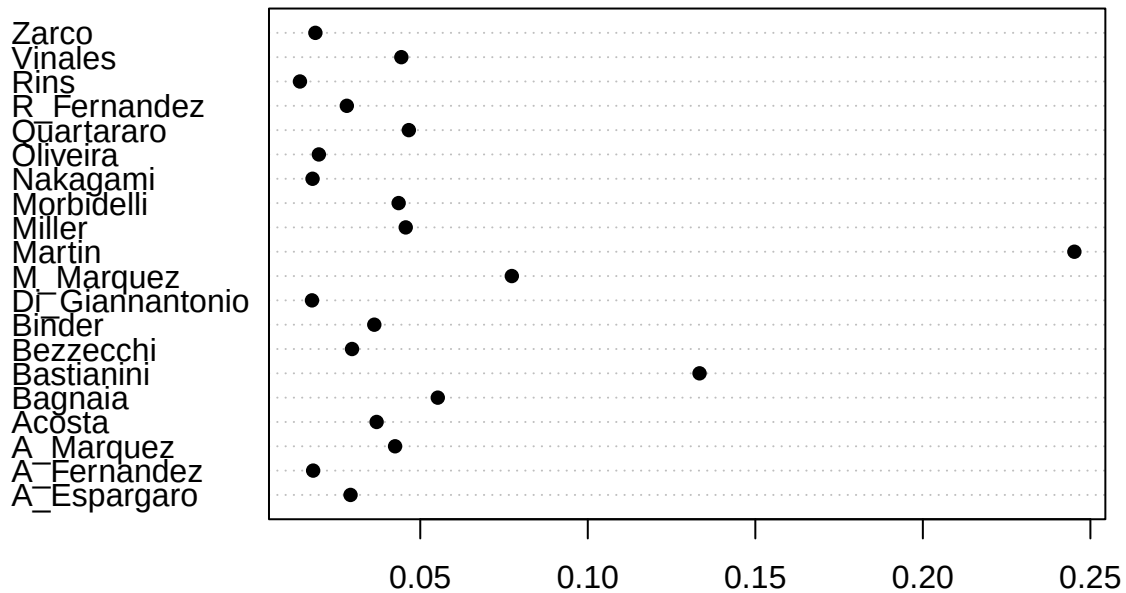
## log(A_Espargaro^15 * (A_Espargaro + A_Fernandez + A_Marquez + Acosta +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio + M_Marquez
## + Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Rins + Vinales + Zarco)^-20 * (A_Espargaro + A_Fernandez
## + A_Marquez + Acosta + Bagnaia + Bastianini + Bezzecchi + Binder +
## Di_Giannantonio + M_Marquez + Martin + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Zarco)^-2 * (A_Espargaro +
## A_Fernandez + A_Marquez + Acosta + Bagnaia + Bastianini + Bezzecchi +
## Binder + Di_Giannantonio + M_Marquez + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Vinales + Zarco)^-7 *
## (A_Espargaro + A_Fernandez + A_Marquez + Acosta + Bagnaia + Bastianini
## + Bezzecchi + Binder + Di_Giannantonio + M_Marquez + Miller +
## Morbidelli + Nakagami + Quartararo + R_Fernandez + Rins + Vinales +
```

```
## Zarco)^-1 * (A_Espargaro + A_Fernandez + A_Marquez + Acosta + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Giannantonio + Martin + Miller +
## Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez + Rins +
## Vinales + Zarco)^-1)
```

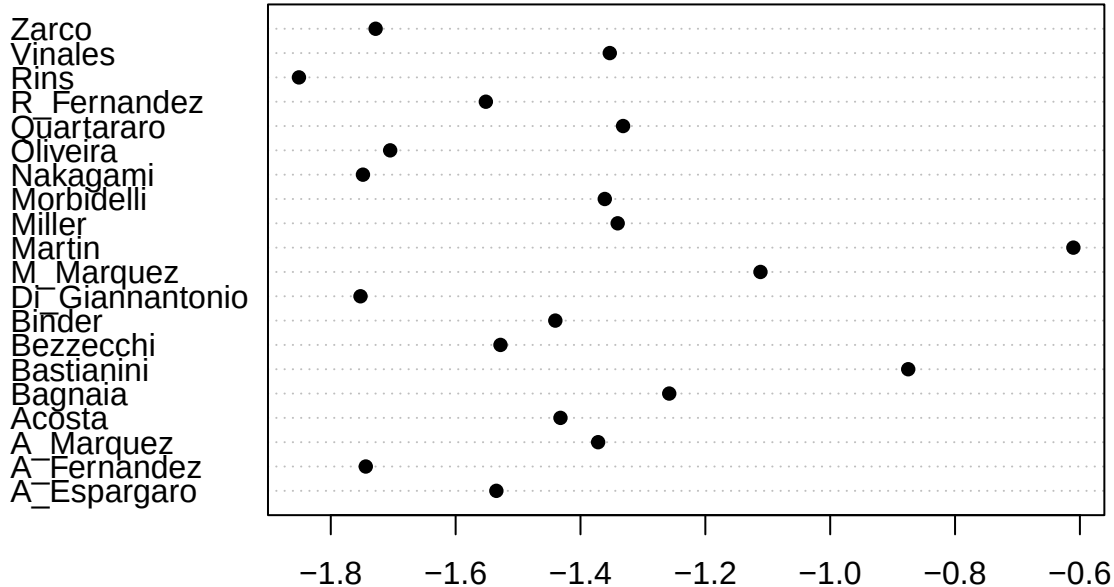
```
sprint_maxp24 <- maxp(L_Sprint24)
sprint_maxp24
```

##	A_Espargaro	A_Fernandez	A_Marquez	Acosta	Bagnaia
##	0.02918845	0.01803176	0.04248592	0.03697638	0.05522831
##	Bastianini	Bezzecchi	Binder	Di_Giannantonio	M_Marquez
##	0.13335109	0.02963763	0.03628169	0.01768487	0.07732259
##	Martin	Miller	Morbidelli	Nakagami	Oliveira
##	0.24522289	0.04566253	0.04354397	0.01784477	0.01972996
##	Quartararo	R_Fernandez	Rins	Vinales	Zarco
##	0.04658412	0.02807886	0.01409309	0.04435250	0.01869861

```
dotchart(sprint_maxp24,pch=16)
```



```
dotchart(log10(sprint_maxp24),pch=16)
```



```
Race24 <- read.table("Race_2024.txt")
```

```
L_Race24 <- ordertable2supp(Race24[1:20,])
```

```
head(L_Race24)
```

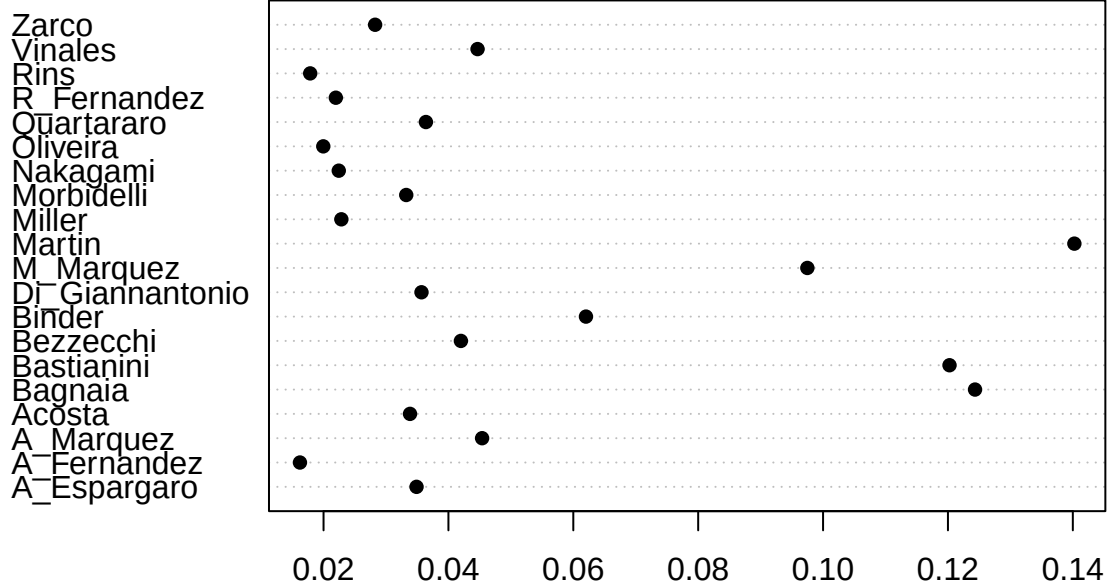
```
## Warning in print.hyper2(x): powers have nonzero sum
```

```
## log(A_Espargaro^15 * (A_Espargaro + A_Fernandez + A_Marquez + Acosta +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Giannantonio + M_Marquez
## + Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Rins + Vinales + Zarco)^-20 * (A_Espargaro + A_Fernandez
## + A_Marquez + Acosta + Bagnaia + Bastianini + Bezzecchi + Binder +
## Di_Giannantonio + M_Marquez + Martin + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Zarco)^-1 * (A_Espargaro +
## A_Fernandez + A_Marquez + Acosta + Bagnaia + Bastianini + Bezzecchi +
## Binder + Di_Giannantonio + M_Marquez + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Vinales + Zarco)^-3 *
## (A_Espargaro + A_Fernandez + A_Marquez + Acosta + Bagnaia + Bastianini
## + Bezzecchi + Binder + Di_Giannantonio + Martin + Miller + Morbidelli +
## Nakagami + Oliveira + Quartararo + R_Fernandez + Rins + Vinales +
## Zarco)^-3 * (A_Espargaro + A_Fernandez + A_Marquez + Acosta + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Giannantonio + Miller + Morbidelli
## + Nakagami + Oliveira + Quartararo + R_Fernandez + Rins + Vinales +
## Zarco)^-3)
```

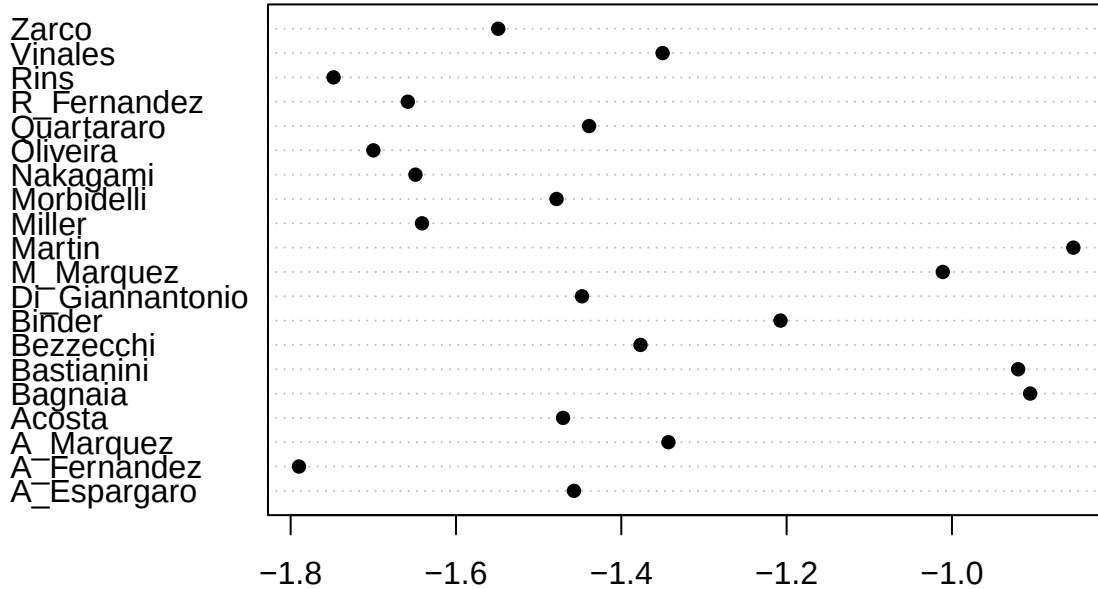
```
race_maxp24 <- maxp(L_Race24)
race_maxp24
```

##	A_Espargaro	A_Fernandez	A_Marquez	Acosta	Bagnaia
##	0.03489107	0.01621965	0.04539233	0.03384698	0.12433704
##	Bastianini	Bezzecchi	Binder	Di_Giannantonio	M_Marquez
##	0.12025008	0.04199902	0.06202624	0.03568326	0.09748150
##	Martin	Miller	Morbidelli	Nakagami	Oliveira
##	0.14024732	0.02285039	0.03324110	0.02243685	0.01995699
##	Quartararo	R_Fernandez	Rins	Vinales	Zarco
##	0.03639186	0.02196670	0.01786178	0.04466378	0.02825606

```
dotchart(race_maxp24,pch=16)
```



```
dotchart(log10(race_maxp24),pch=16)
```



```
##MotoGP2023
```

```
head(MotoGP2023_Sprint)
```

```
## # A tibble: 6 x 23
##   Rider Bike Team S.POR S.ARG S.AME S.SPA S.FRA S.ITA S.GER S.NED S.GBR S.AUT
##   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Bagna~ Duca~ Duca~ 1     6     1     2     3     1     2     2    14     1
## 2 Martin Duca~ Prim~ 2     8     3     4     1     3     1     6     6     3
## 3 Bezze~ Duca~ Moon~ Ret    2     6     9     7     2     7     1     2    Ret
## 4 Binder KTM   RedB~ 12    1     5     1     2    11     6     5     9     2
## 5 Zarco Duca~ Prim~ 8     12    11     8     6     4     5    13     4    Ret
## 6 A.Esp~ Apri~ Apri~ 6     Ret    4     Ret    8     8     9     4     5     7
## # i 10 more variables: S.CAT <chr>, S.RSM <chr>, S.IND <chr>, S.JPN <chr>,
## #   S.INA <chr>, S.AUS <chr>, S.THA <chr>, S.MAL <chr>, S.QAT <chr>,
## #   S.VAL <chr>
```

```
top5_sprint23 <- MotoGP2023_Sprint %>%
  slice(1:5) %>%
  pivot_longer(
    cols = S.POR:S.VAL,
    names_to = "Sprint_Race23",
    values_to = "Result"
  ) %>%
  mutate(
```

```

    Position = as.numeric(Result),
    Sprint_Race23 = factor(Sprint_Race23, levels = names(MotoGP2023_Sprint)[4:23])
  )

```

```

## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Position = as.numeric(Result)'.
## Caused by warning:
## ! NA introdotti per coercizione

```

```

ggplot(top5_sprint23, aes(x = Sprint_Race23, y = Position, colour = Rider, group = Rider)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_reverse() +
  labs(
    title = "Top 5 Riders Sprint Race Performance 2023",
    x = "Sprint Race",
    y = "Finishing Position",
    colour = "Rider"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

```

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_line()').

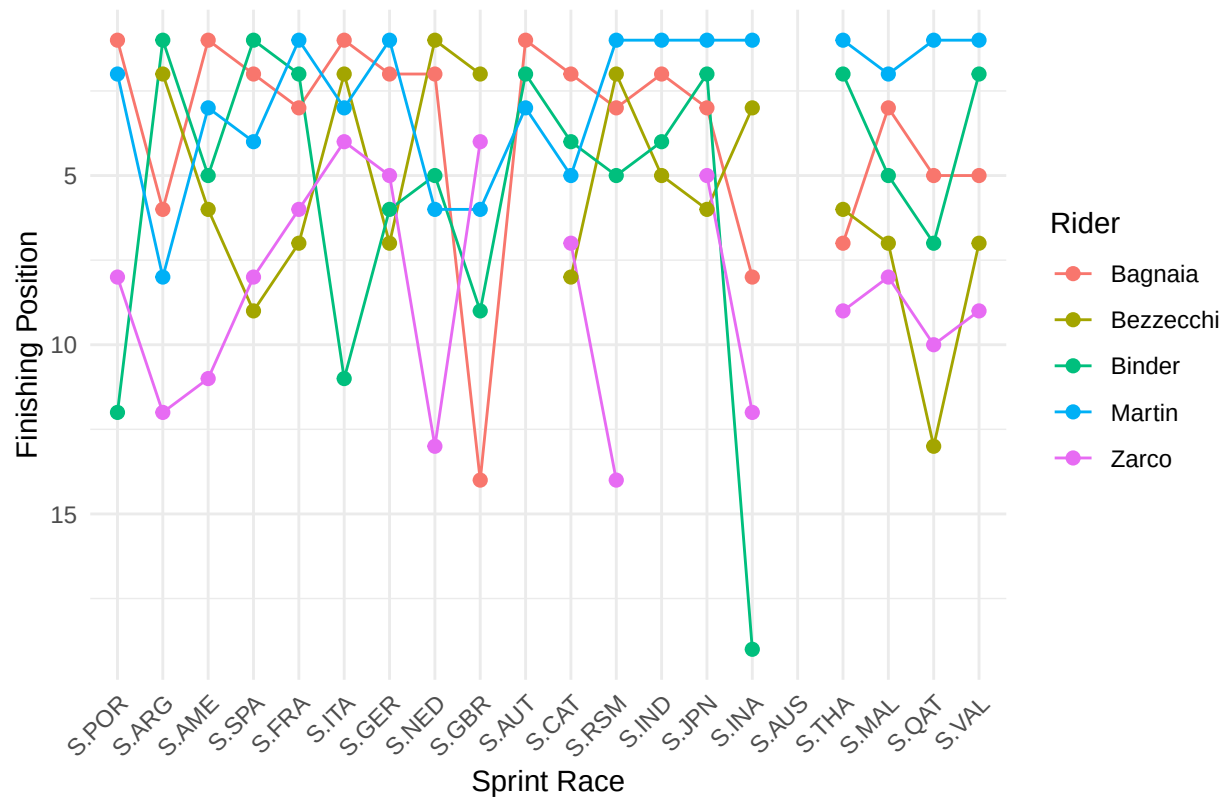
```

```

## Warning: Removed 9 rows containing missing values or values outside the scale range
## ('geom_point()').

```

Top 5 Riders Sprint Race Performance 2023



```
head(MotoGP2023_Race)
```

```
## # A tibble: 6 x 23
##   Rider Bike Team POR ARG AME SPA FRA ITA GER NED GBR AUT
##   <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr> <chr>
## 1 Bagna~ Duca~ Duca~ 1 16 Ret 1 Ret 1 2 1 2 1
## 2 Martin Duca~ Prim~ Ret 5 Ret 4 2 2 1 5 6 7
## 3 Bezze~ Duca~ Moon~ 3 1 6 Ret 1 8 4 2 Ret 3
## 4 Binder KTM RedB~ 6 17 13 2 6 5 Ret 4 3 2
## 5 Zarco Duca~ Prim~ 4 2 7 Ret 3 3 3 Ret 9 13
## 6 A.Esp~ Apri~ Apri~ 9 15 Ret 5 5 6 16 3 1 9
## # i 10 more variables: CAT <chr>, RSM <chr>, IND <chr>, JPN <chr>, INA <chr>,
## # AUS <chr>, THA <chr>, MAL <chr>, QAT <chr>, VAL <chr>
```

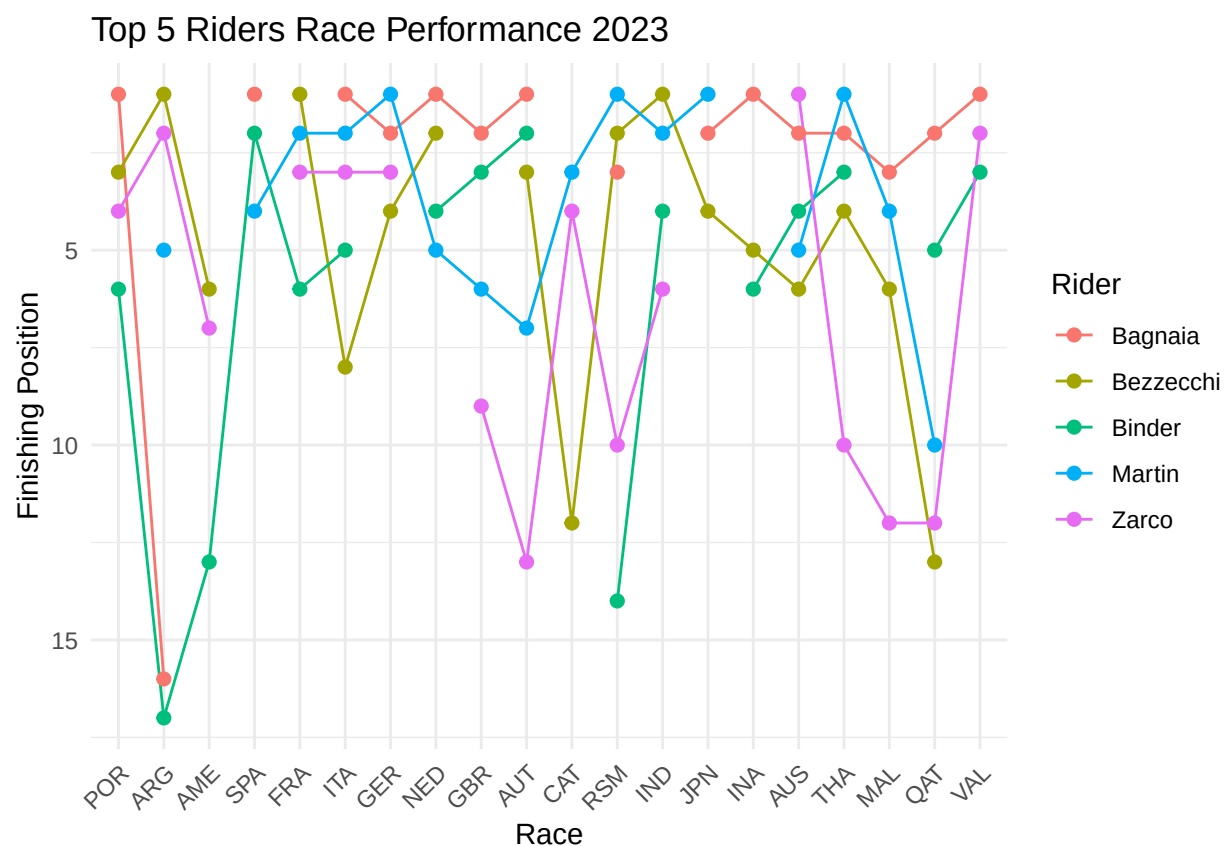
```
top5_race23 <- MotoGP2023_Race %>%
  slice(1:5) %>%
  pivot_longer(
    cols = POR:VAL,
    names_to = "Race23",
    values_to = "Result"
  ) %>%
  mutate(
    Position = as.numeric(Result),
    Race23 = factor(Race23, levels = names(MotoGP2023_Race)[4:23])
  )
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Position = as.numeric(Result)'.
## Caused by warning:
## ! NA introdotti per coercizione
```

```
ggplot(top5_race23, aes(x = Race23, y = Position, colour = Rider, group = Rider)) +
  geom_line() +
  geom_point(size = 2) +
  scale_y_reverse() +
  labs(
    title = "Top 5 Riders Race Performance 2023",
    x = "Race",
    y = "Finishing Position",
    colour = "Rider"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
## ('geom_line()').
```

```
## Warning: Removed 19 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
moto_2023<-read.table("GP_2023.txt")
```

```
L2023<- ordertable2supp(moto_2023[1:20,-ncol(moto_2023)])
head(L2023)
```

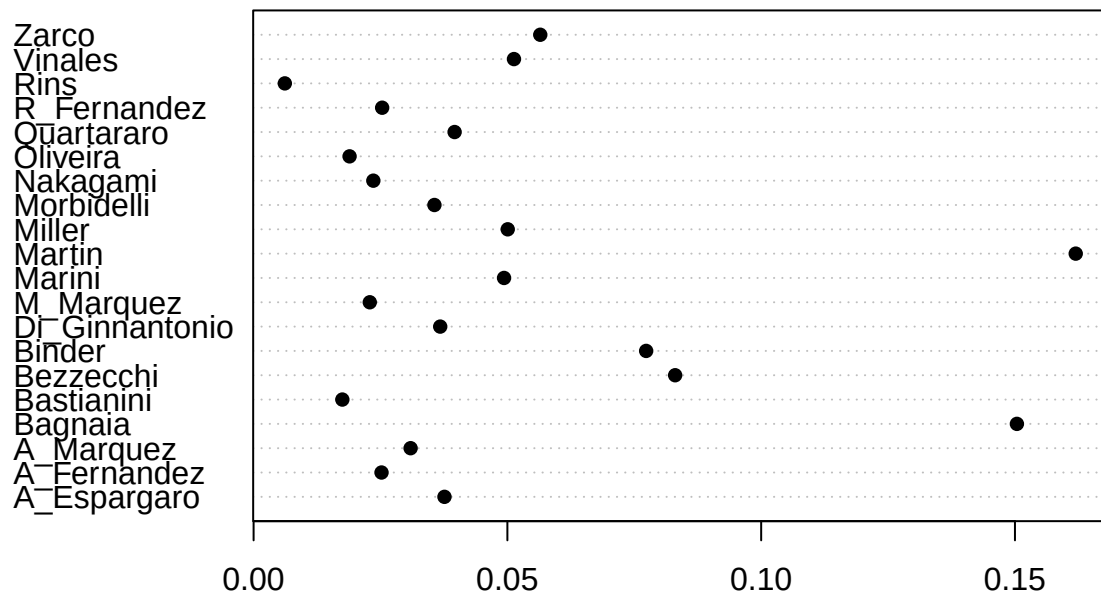
```
## Warning in print.hyper2(x): powers have nonzero sum
```

```
## log(A_Espargaro^29 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini +
## Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Rins + Vinales)^-1 * (A_Espargaro + A_Fernandez +
## A_Marquez + Bagnaia + Bastianini + Bezzecchi + Binder + Di_Ginnantonio
## + M_Marquez + Marini + Martin + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Vinales + Zarco)^-39 *
## (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia + Bastianini +
## Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini + Martin +
## Miller + Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez +
## Vinales + Zarco)^-1 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia
## + Bastianini + Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini
## + Miller + Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez
## + Rins + Vinales + Zarco)^-13 * (A_Espargaro + A_Fernandez + A_Marquez
## + Bagnaia + Bastianini + Bezzecchi + Binder + Di_Ginnantonio +
## M_Marquez + Martin + Miller + Morbidelli + Nakagami + Oliveira +
## Quartararo + R_Fernandez + Vinales + Zarco)^-1)
```

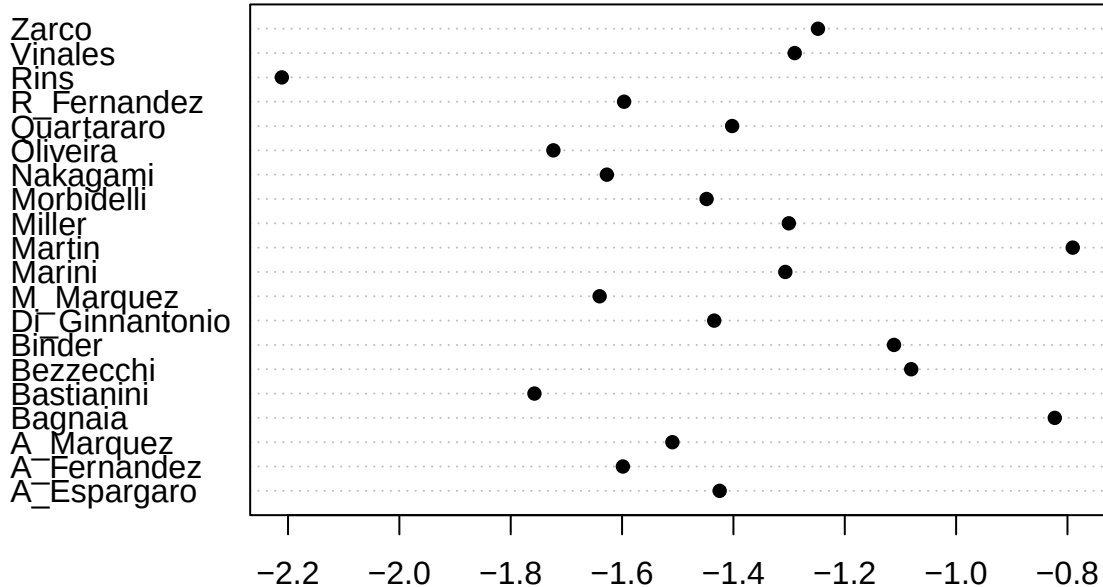
```
moto_maxp23 <- maxp(L2023)
moto_maxp23
```

##	A_Espargaro	A_Fernandez	A_Marquez	Bagnaia	Bastianini
##	0.037604649	0.025203207	0.030931506	0.150383099	0.017481082
##	Bezzecchi	Binder	Di_Ginnantonio	M_Marquez	Marini
##	0.083050800	0.077339176	0.036766899	0.022890917	0.049342197
##	Martin	Miller	Morbidelli	Nakagami	Oliveira
##	0.161971283	0.050066760	0.035626084	0.023587259	0.018907737
##	Quartararo	R_Fernandez	Rins	Vinales	Zarco
##	0.039599175	0.025328628	0.006149497	0.051295578	0.056474467

```
dotchart(moto_maxp23,pch=16)
```



```
dotchart(log10(moto_maxp23),pch=16)
```



```
Sprint23 <-read.table("Sprint_2023.txt")

L_Sprint23 <- ordertable2supp(Sprint23[1:20,])
head(L_Sprint23)
```

Difference in Rider Performance Between Sprint and Race 2023

```
## Warning in print.hyper2(x): powers have nonzero sum

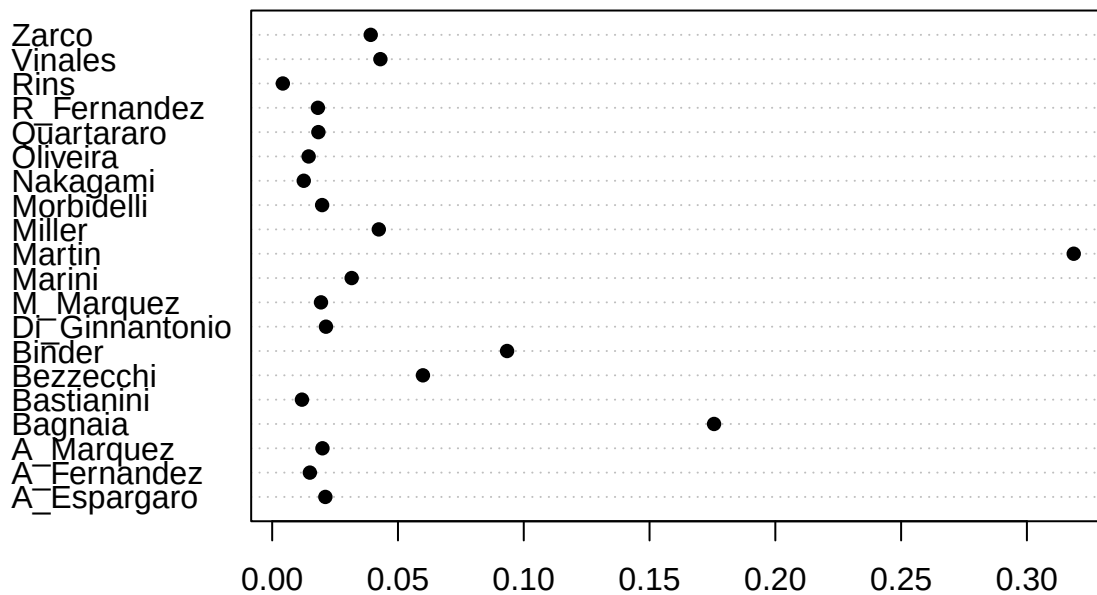
## log(A_Espargaro^13 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini +
## Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Rins + Vinales + Zarco)^-19 * (A_Espargaro + A_Fernandez
## + A_Marquez + Bagnaia + Bastianini + Bezzecchi + Binder +
## Di_Ginnantonio + M_Marquez + Marini + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Vinales + Zarco)^-9 *
## (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia + Bastianini +
## Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Miller + Morbidelli +
## Nakagami + Oliveira + Quartararo + R_Fernandez + Rins + Vinales +
## Zarco)^-1 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia +
## Bastianini + Bezzecchi + Binder + M_Marquez + Marini + Miller +
## Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez + Rins +
```

```
## Vinales + Zarco)^-1 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia
## + Bastianini + Bezzecchi + Binder + M_Marquez + Miller + Morbidelli +
## Nakagami + Oliveira + Quartararo + R_Fernandez + Rins + Vinales +
## Zarco)^-1)
```

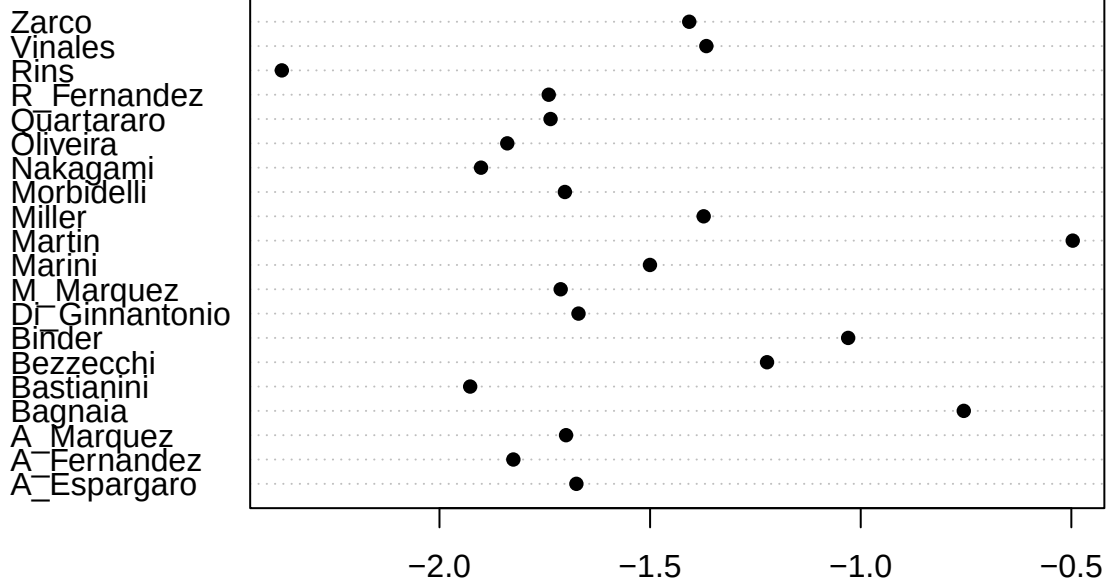
```
sprint_maxp23 <- maxp(L_Sprint23)
sprint_maxp23
```

##	A_Espargaro	A_Fernandez	A_Marquez	Bagnaia	Bastianini
##	0.021135335	0.014975391	0.019975447	0.175643228	0.011826230
##	Bezzecchi	Binder	Di_Ginnantonio	M_Marquez	Marini
##	0.059917092	0.093362872	0.021381105	0.019396051	0.031597933
##	Martin	Miller	Morbidelli	Nakagami	Oliveira
##	0.318606276	0.042379258	0.019847749	0.012548852	0.014486268
##	Quartararo	R_Fernandez	Rins	Vinales	Zarco
##	0.018351598	0.018170175	0.004224186	0.043008866	0.039166088

```
dotchart(sprint_maxp23,pch=16)
```



```
dotchart(log10(sprint_maxp23),pch=16)
```



```
Race23 <- read.table("Race_2023.txt")
```

```
L_Race23 <- ordertable2supp(Race23[1:20,])
```

```
head(L_Race23)
```

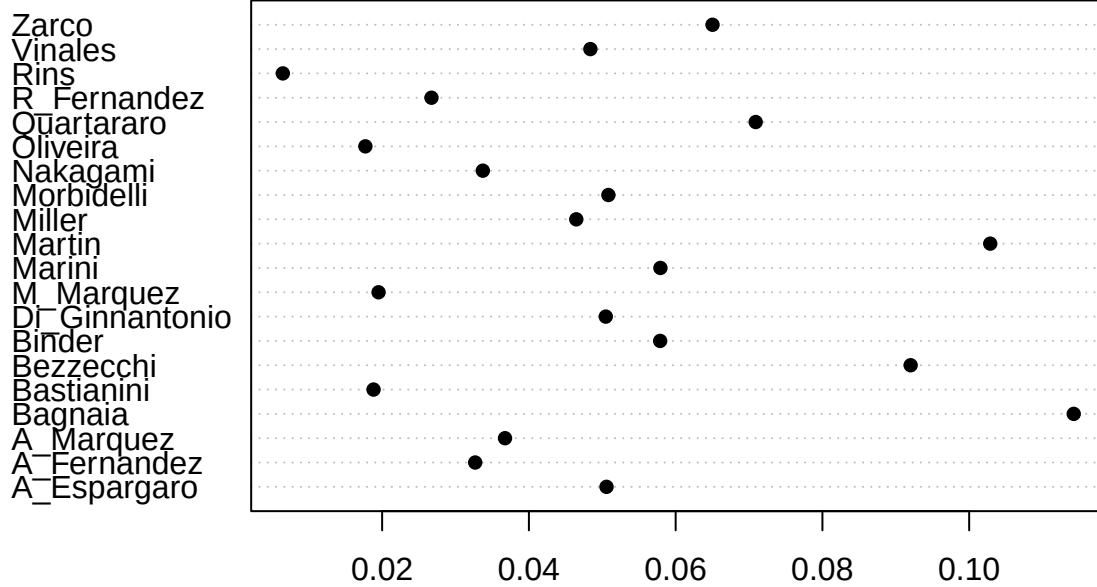
```
## Warning in print.hyper2(x): powers have nonzero sum
```

```
## log(A_Espargaro^16 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia +
## Bastianini + Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini +
## Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Rins + Vinales)^-1 * (A_Espargaro + A_Fernandez +
## A_Marquez + Bagnaia + Bastianini + Bezzecchi + Binder + Di_Ginnantonio
## + M_Marquez + Marini + Martin + Miller + Morbidelli + Nakagami +
## Oliveira + Quartararo + R_Fernandez + Rins + Vinales + Zarco)^-20 *
## (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia + Bastianini +
## Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini + Martin +
## Miller + Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez +
## Vinales + Zarco)^-1 * (A_Espargaro + A_Fernandez + A_Marquez + Bagnaia
## + Bastianini + Bezzecchi + Binder + Di_Ginnantonio + M_Marquez + Marini
## + Miller + Morbidelli + Nakagami + Oliveira + Quartararo + R_Fernandez
## + Rins + Vinales + Zarco)^-4 * (A_Espargaro + A_Fernandez + A_Marquez +
## Bagnaia + Bastianini + Bezzecchi + Binder + Di_Ginnantonio + M_Marquez
## + Martin + Miller + Morbidelli + Nakagami + Oliveira + Quartararo +
## R_Fernandez + Vinales + Zarco)^-1)
```

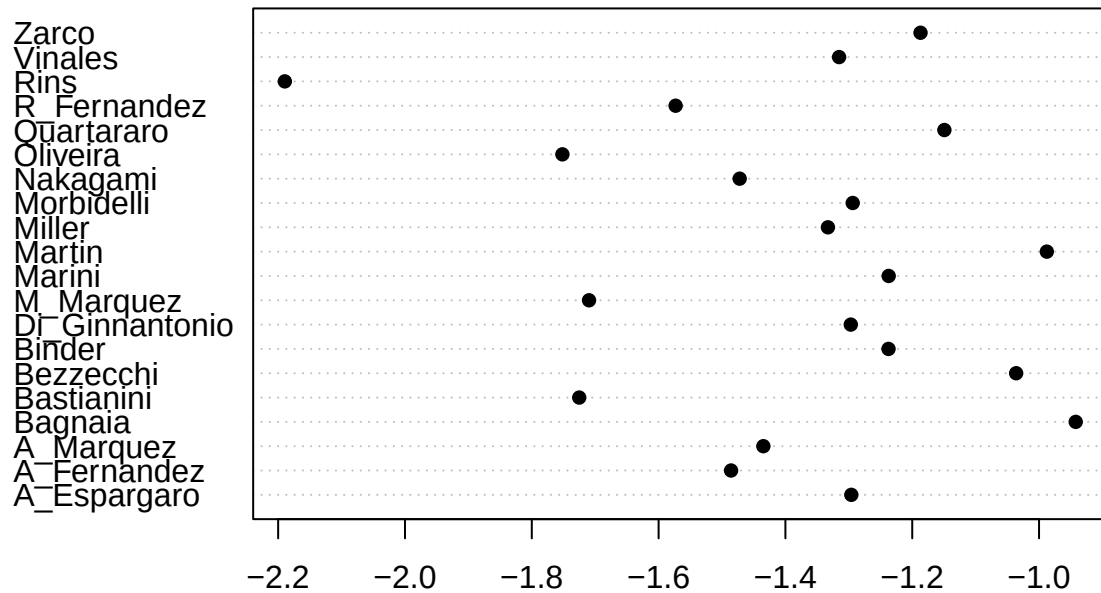
```
race_maxp23 <- maxp(L_Race23)
race_maxp23
```

```
##      A_Espargaro      A_Fernandez      A_Marquez      Bagnaia      Bastianini
##      0.050577111      0.032681020      0.036745132      0.114248043      0.018827636
##      Bezzecchi      Binder Di_Ginnantonio      M_Marquez      Marini
##      0.092022263      0.057888100      0.050477551      0.019511953      0.057924937
##      Martin      Miller      Morbidelli      Nakagami      Oliveira
##      0.102871338      0.046457949      0.050833360      0.033719485      0.017712927
##      Quartararo      R_Fernandez      Rins      Vinales      Zarco
##      0.070910093      0.026720888      0.006463996      0.048379751      0.065026465
```

```
dotchart(race_maxp23,pch=16)
```



```
dotchart(log10(race_maxp23),pch=16)
```



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.